

Functional Analysis — Homework 4

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Problem 1. Fix $1 \leq p \leq \infty$ and let $S_L : \ell^p \rightarrow \ell^p$ be the left-shift operator:

$$S_L(x_1, x_2, x_3, \dots) = S(x_2, x_3, x_4, \dots).$$

Show that S_L is bounded and compute its operator norm. Does the equation $S_L(x) = x$ have any solutions other than $x = 0$?

Proof. Let $(x_n) \in \ell^p$ such that $\|(x_n)\|_p = 1$. First assume $p < \infty$. We have

$$\begin{aligned} \|S_L((x_n))\|_p^p &= \sum_{n=2}^{\infty} |x_n|^p \\ &= \left(\sum_{n=1}^{\infty} |x_n|^p \right) - |x_1|^p \\ &= \|(x_n)\|_p^p - |x_1|^p \\ &= 1 - |x_1|^p \\ &\leq 1, \end{aligned}$$

so S_L is bounded. In fact, this upper bound of 1 becomes equality whenever $x_1 = 0$, so $\|S_L\|_{op} = 1$. We note that for $S_L(x) = x$ to be satisfied, we must have $x_i = x_{i+1}$ for all $i \in \mathbb{Z}^+$, so in other words x must be a constant sequence. In this case (that is, still assuming $p < \infty$), all elements of ℓ^p must be sequences converging to 0, so since the only such constant sequence is the zero sequence, it is indeed the only solution to $S_L(x) = x$.

Now if $p = \infty$, then if $\|(x_n)\|_{\infty} = 1$,

$$\begin{aligned} \|S_L((x_n))\|_{\infty} &= \sup_{n \geq 2} |x_n| \\ &\leq \sup_{n \geq 1} |x_n| \\ &= \|(x_n)\|_{\infty} \\ &= 1, \end{aligned}$$

so S_L is bounded. Again, equality is achieved when $x_1 = 0$, so we have $\|S_L\|_{op} = 1$. Any solution to $S_L(x) = x$ must still be a constant sequence by the same reasoning as above, but in this case not all elements of ℓ^{∞} are sequences converging to 0; they just need to be termwise bounded from above. This is satisfied by any constant sequence, so in fact there are infinitely many solutions to $S_L(x) = x$ in ℓ^{∞} , namely all constant sequences. $\square$

Problem 2. Show that the closed unit ball of c_0 , which is the set

$$B_1[0] = \{x \in c_0 : \|x\|_\infty \leq 1\},$$

is a closed convex set with no extreme points.

Proof. Since norms are continuous, this is the preimage of the closed set $[0, 1]$ under the function $x \mapsto \|x\|_\infty$, so $B_1[0]$ is closed by continuity. For convexity, let $x, y \in B_1[0]$. We want to show that $tx + (1 - t)y \in B_1[0]$ whenever $0 \leq t \leq 1$. Indeed,

$$\begin{aligned} \|tx + (1 - t)y\|_\infty &\leq |t| \cdot \|x\|_\infty + |1 - t| \cdot \|y\|_\infty \\ &= t \cdot \|x\|_\infty + (1 - t) \cdot \|y\|_\infty \\ &\leq t + (1 - t) \\ &= 1, \end{aligned}$$

where the third line follows from $\|x\|_\infty \leq 1$ and $\|y\|_\infty \leq 1$. Thus $B_1[0]$ is convex.

Now let $z \in B_1[0]$. We want to find $x \neq y \in B_1[0]$ and $t \in [0, 1]$ such that $tx + (1 - t)y = z$. Since z converges to zero, we can choose $N \in \mathbb{Z}^+$ such that when $n \geq N$, $|z_n| \leq 1/2$. This allows us to define x to be equal to z except that $x_N = 2z_N$ and $x_{N+1} = 0$, and we can also define y to be equal to z except that $y_N = 0$ and $y_{N+1} = 2z_{N+1}$. Both x and y are in $B_1[0]$ since $|2z_N|, |2z_{N+1}| \leq 1$ by construction, and we have

$$\frac{1}{2}x_i + \frac{1}{2}y_i = \begin{cases} \frac{1}{2}2z_i + \frac{1}{2}0 = z_i & i = N \\ \frac{1}{2}0 + \frac{1}{2}2z_i = z_i & i = N + 1 \\ \frac{1}{2}z_i + \frac{1}{2}z_i = z_i & \text{otherwise,} \end{cases}$$

so $\frac{1}{2}x + \frac{1}{2}y = z$ and $x \neq y \in B_1[0]$ as desired. $\square$

Problem 3. Recall that $C^0[0, 1]$ is the space of continuous functions $f : [0, 1] \rightarrow \mathbb{R}$ equipped with the sup norm

$$\|f\|_\infty = \sup_{x \in [0, 1]} |f(x)|.$$

Find the extreme points of the closed unit ball $B_1[0]$ of this space. Conclude that $B_1[0]$ is **not** the closure of the convex hull of its extreme points (it does not satisfy the conclusion of Krein-Milman's theorem).

Proof. We want to find $f \in C^0[0, 1]$ such that $f = tg + (1 - t)h$ for some $g, h \in C^0[0, 1]$ and $t \in [0, 1]$ implies that $g = h = f$. We begin by applying the condition to a single point x such that $f(x) = \sup |f| = \|f\|_\infty$ (existence is guaranteed because f is continuous and thus achieves a maximum on the compact set $[0, 1]$). If $|f(x)| < 1$, then we can choose an $\varepsilon > 0$ such that $|f(x) + \varepsilon| < 1$ and $|f(x) - \varepsilon| < 1$. We can then choose $g = f + \varepsilon$ and $h = f - \varepsilon$ (both in $B_1[0]$ by construction), so that $f = \frac{1}{2}g + \frac{1}{2}h$, and since $g \neq h$, this shows that f is not an extreme point. Thus for any f which *is* an extreme point, we must have $\|f\|_\infty = 1$.

I can't figure out how to write down the next part, but the previous argument holds when $f(x) < 1$ for any x , not just when $f(x) = \|f\|_\infty$, if instead of doing a $\pm\varepsilon$ you do something

like scaling it just a little bit so that the function still stays within $\|f\|_\infty \leq 1$ but is nudged up a bit whenever $f(x) \neq \|f\|_\infty$ for g and then h is the “nudged/scaled down” version.

Thus we must have that $|f(x)| = 1$ for all x if f is an extreme point, so the extreme points are the constant functions at -1 and 1 (call them g and h respectively). Only constant functions are represented in $[g, h]$, and taking a closure doesn't help us since a sequence of constant functions converges to another constant function if at all, so since there are certainly non-constant functions in $B_1[0]$, we conclude that it is not the closure of the convex hull of its extreme points. $\square$

Problem 4. Let X be a vector space equipped with two norms, call them $\|\cdot\|_1$ and $\|\cdot\|_2$, such that X is Banach with respect to either of these two norms. Show that if these norms are *comparable*, i.e. if there exists a constant $C > 0$ such that

$$\|x\|_1 \leq C\|x\|_2, \quad \forall x \in X,$$

then they are *equivalent*, i.e. there also exists another constant $C' > 0$ such that

$$\|x\|_2 \leq C'\|x\|_1, \quad \forall x \in X.$$

Proof. Consider the inclusion map $i : X_1 \rightarrow X_2$, where X_i is X equipped with $\|\cdot\|_i$. This is bounded by assumption that $\|x\|_1 \leq C\|x\|_2$ for some $C > 0$, and it is naturally bijective, so the open mapping theorem tells us that it has a continuous inverse (this is the fact that i is an open map together with bijectivity, since the preimage of open sets over i^{-1} is just the image of open sets under i). Thus i^{-1} is bounded and the desired constant $C' > 0$ exists. $\square$

Problem 5. Let X be a Banach space and $Y, Z \subset X$ be closed subspaces such that

$$(1) \ Y \cap Z = \{0\}.$$

$$(2) \ Y + Z = X \text{ (every } x \in X \text{ can be written as } x = y + z \text{ with } y \in Y, z \in Z\text{)}.$$

Show that X is isomorphic to $Y \oplus Z$. In this situation, we say that Y (and by symmetry also Z) is a **complemented** subspace of X .

Proof. First note that if $y_1 + z_1 = y_2 + z_2$, we have $y_1 - y_2 = z_2 - z_1$, where the left side lies in Y and the right side lies in Z , so the two sides must be in $Y \cap Z = \{0\}$, implying $y_1 = y_2$ and $z_1 = z_2$. We are then justified in defining $\varphi : Y + Z \rightarrow Y \oplus Z$ by $y + z \mapsto (y, z)$. This is linear since

$$\begin{aligned} \varphi(a(y_1 + z_1) + b(y_2 + z_2)) &= \varphi(ay_1 + by_2 + az_1 + bz_2) \\ &= (ay_1 + by_2, az_1 + bz_2) \\ &= (ay_1, az_1) + (by_2, bz_2) \\ &= a(y_1, z_1) + b(y_2, z_2) \\ &= a\varphi(y_1 + z_1) + b\varphi(y_2 + z_2). \end{aligned}$$

If $\varphi(y_1 + z_1) = \varphi(y_2 + z_2)$, then $(y_1, z_1) = (y_2, z_2)$, so $y_1 = y_2$ and $z_1 = z_2$, showing that φ is injective. Surjectivity comes easily as well; for any $(y, z) \in Y \oplus Z$, $\varphi(y + z)$.

For boundedness of the inverse, we get

$$\begin{aligned}\frac{\|\varphi^{-1}(y, z)\|}{\|(y, z)\|} &= \frac{\|y + z\|}{\|y\| + \|z\|} \\ &\leq \frac{\|y\| + \|z\|}{\|y\| + \|z\|} \\ &= 1\end{aligned}$$

whenever $(y, z) \neq (0, 0)$. Boundedness of φ itself is the one part I haven't been able to figure out. $\square$

Problem 6. Let X be a normed space. In the language of the previous problem, show that any *finite-dimensional* subspace $Y \subset X$ is complemented.

Proof. Let $B = \{y_1, \dots, y_n\}$ be a basis of Y , extend to a Hamel basis, and let $C = \{z_i\}_{i \in I}$ be the remainder of the extended basis. Let Z be the space spanned by C . Clearly $Z \cap Y = \{0\}$, and every $x \in X$ can be written as a linear combination of elements from B and C by definition of a Hamel basis, so we just need to show that Z is closed.

Let x be a limit point of Z and (x_n) a sequence in Z converging to x . Then x has a unique representation as $y + z$, where $y \in Y$ and $z \in Z$ (uniqueness is given by the first sentence of the proof of the Problem 5 since it makes no use of closure). $\square$